

créer les dossiers:

```
C:\Users\karab\OneDrive\Documents\cours bachelor\conteneurisation\tp2-docker>mkdir frontend backend database

C:\Users\karab\OneDrive\Documents\cours bachelor\conteneurisation\tp2-docker>dir
Le volume dans le lecteur C s'appelle Windows
Le numéro de série du volume est 0461-2F70

Répertoire de C:\Users\karab\OneDrive\Documents\cours bachelor\conteneurisation\tp2-docker

20/05/2026  12:31    <DIR>        .
20/05/2026  12:26    <DIR>        ..
20/05/2026  12:31    <DIR>        backend
20/05/2026  12:31    <DIR>        database
20/05/2026  11:39                74 Dockerfile
20/05/2026  12:31    <DIR>        frontend
20/05/2026  11:37    <DIR>        site
                1 fichier(s)                74 octets
                6 Rép(s) 118 263 943 168 octets libres

C:\Users\karab\OneDrive\Documents\cours bachelor\conteneurisation\tp2-docker>
```

on creer les pages:

```
C:\Users\karab\OneDrive\Documents\cours bachelor\conteneurisation\tp2-docker>cd frontend
C:\Users\karab\OneDrive\Documents\cours bachelor\conteneurisation\tp2-docker\frontend>type nul > Dockerfile
C:\Users\karab\OneDrive\Documents\cours bachelor\conteneurisation\tp2-docker\frontend>type nul > index.html
C:\Users\karab\OneDrive\Documents\cours bachelor\conteneurisation\tp2-docker\frontend>
```

dans dockerfile:

```
Fichier    Modifier    Affichage    H

FROM nginx:alpine

COPY index.html /usr/share/nginx/html/

EXPOSE 80
```

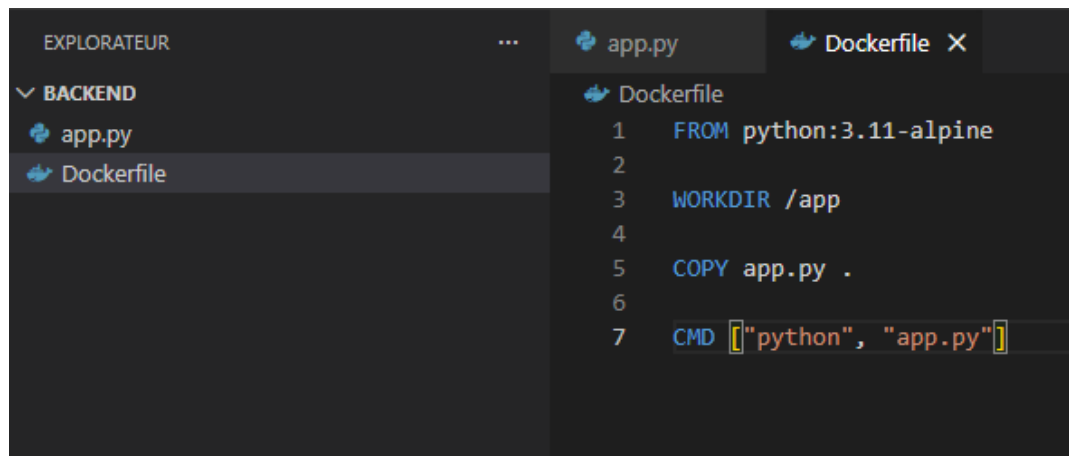
dans index.html:

```
C: > Users > karab > OneDrive > Documents > cours bachelor > conteneurisation
1  <h1>Frontend Docker</h1>
```

on creer cotes serveur les fichiers:

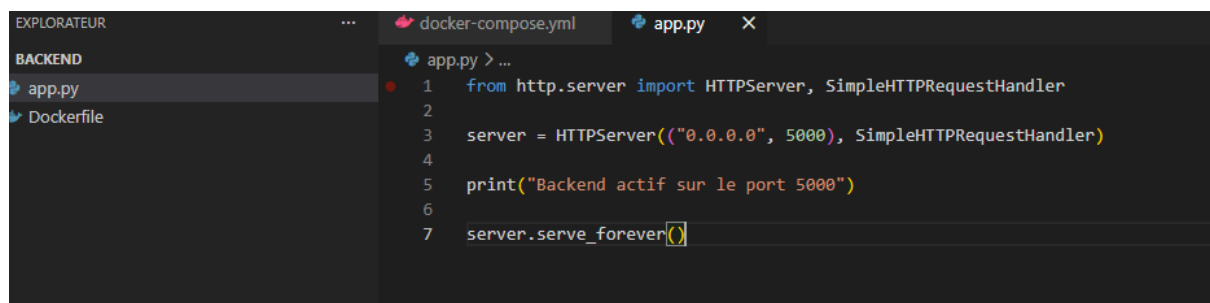
```
C:\Users\karab\OneDrive\Documents\cours bachelor\conteneurisation\tp2-docker\backend>type nul > Dockerfile
C:\Users\karab\OneDrive\Documents\cours bachelor\conteneurisation\tp2-docker\backend>type nul > app.py
```

on modifie les fichier cotes backend:



The screenshot shows a code editor with a sidebar on the left labeled 'EXPLORATEUR' containing a folder 'BACKEND' with files 'app.py' and 'Dockerfile'. The main editor area has two tabs: 'app.py' and 'Dockerfile'. The 'Dockerfile' tab is active, displaying the following content:

```
Dockerfile
1 FROM python:3.11-alpine
2
3 WORKDIR /app
4
5 COPY app.py .
6
7 CMD ["python", "app.py"]
```



The screenshot shows the same code editor with the 'app.py' tab active. The sidebar now shows 'app.py' and 'Dockerfile' under the 'BACKEND' folder. The main editor area displays the following Python code:

```
app.py > ...
1 from http.server import HTTPServer, SimpleHTTPRequestHandler
2
3 server = HTTPServer(("0.0.0.0", 5000), SimpleHTTPRequestHandler)
4
5 print("Backend actif sur le port 5000")
6
7 server.serve_forever()
```

créer le docker compose :

- type nul > docker-compose.yml

```
docker-compose.yml X app.py
C: > Users > karab > OneDrive > Documents > cours bachelor > conteneurisation > tp2-docker > docker-compose.yml
1  services:
2
3  frontend:
4    image: nginx:alpine
5    container_name: frontend
6    ports:
7      - "8080:80"
8    volumes:
9      - ./frontend/index.html:/usr/share/nginx/html/index.html
10   networks:
11     - app-network
12
13   backend:
14     build: ./backend
15     container_name: backend
16     networks:
17       - app-network
18
19   db:
20     image: mysql:8.0
21     container_name: mysql-db
22     restart: always
23     environment:
24       MYSQL_ROOT_PASSWORD: root
25       MYSQL_DATABASE: appdb
26     volumes:
27       - mysql-data:/var/lib/mysql
28     networks:
29       - app-network
30
31   volumes:
32     mysql-data:
33
34   networks:
35     app-network:
```

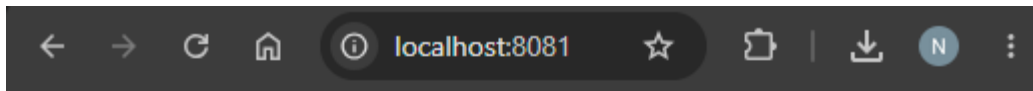
Voulez-vous installer Microsoft recommar

lancer le projet:

- docker compose up -d

```
C:\Users\karab\OneDrive\Documents\cours bachelor\conteneurisation\tp2-docker>docker compose ps
NAME                IMAGE                COMMAND                SERVICE    CREATED        STATUS        PORTS
backend             tp2-docker-backend  "python app.py"        backend    4 seconds ago  Up 4 seconds  0.0.0.0:8081->80/tcp, [::]:8081->80
frontend            tp2-docker-frontend  "/docker-entrypoint.s..." frontend    2 minutes ago  Up 2 minutes  0.0.0.0:8081->80/tcp, [::]:8081->80
mysql-db            mysql:8.0            "docker-entrypoint.s..." db          2 minutes ago  Up 2 minutes  3306/tcp, 33060/tcp
C:\Users\karab\OneDrive\Documents\cours bachelor\conteneurisation\tp2-docker>
```

aller dans localhost:8081



Frontend Docker

les logs :

```
C:\Users\karab\OneDrive\Documents\cours bachelor\conteneurisation\tp2-docker>docker compose logs
mysql-db | 2026-05-20 11:00:14+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.46-1.el9 started.
frontend | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
frontend | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
mysql-db | 2026-05-20 11:00:14+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
mysql-db | 2026-05-20 11:00:14+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.46-1.el9 started.
mysql-db | '/var/lib/mysql/mysql.sock' -> '/var/run/mysqld/mysqld.sock'
frontend | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
frontend | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
frontend | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
mysql-db | 2026-05-20T11:00:15.308456Z 0 [Warning] [MY-011068] [Server] The syntax '--skip-host-cache' is deprecated and will be removed in a future release. Please use SET GLOBAL host_cache_size=0 instead.
mysql-db | 2026-05-20T11:00:15.311966Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 8.0.46) starting as process 1
mysql-db | 2026-05-20T11:00:15.320392Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
mysql-db | 2026-05-20T11:00:15.738369Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
mysql-db | 2026-05-20T11:00:16.056259Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self signed.
mysql-db | 2026-05-20T11:00:16.056309Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS . Encrypted connections are now supported for this channel.
mysql-db | 2026-05-20T11:00:16.060148Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible to all OS users. Consider choosing a different directory.
```

Le problème rencontré venait du port 8080 qui était déjà utilisé par un autre service sur le PC. Nous avons donc changé le port du frontend vers 8081, ce qui m'a permis d'accéder correctement au conteneur nginx.

Conclusion : ce TP nous a permis de découvrir Docker Compose, la gestion d'une architecture multi-conteneurs avec un frontend, un backend et une base de données, ainsi que l'automatisation du déploiement et les bases d'une organisation DevOps.